

Module 11: Genomics and Biotechnology — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain the function of restriction enzymes and DNA ligase in creating recombinant DNA
2. Describe the steps and purpose of PCR (Polymerase Chain Reaction)
3. Explain how gel electrophoresis separates DNA fragments by size
4. Describe how DNA fingerprinting (STR analysis) is used in forensics and paternity testing
5. Outline the process of creating a transgenic organism (GMO)
6. Evaluate the goals, methods, and ethical considerations of gene therapy and CRISPR gene editing

Key Terms to Know

Molecular Tools

- **Restriction Enzyme** — An enzyme that cuts DNA at a specific recognition sequence (e.g., EcoRI cuts GAATTC)
- **Sticky Ends** — Short single-stranded overhangs produced by restriction enzyme cuts; allow compatible fragments to base-pair
- **Blunt Ends** — Straight-cut DNA ends without overhangs
- **DNA Ligase** — Enzyme that permanently joins (seals) DNA fragments by forming phosphodiester bonds
- **Recombinant DNA** — A DNA molecule containing genetic material from two different sources
- **Vector** — A DNA molecule used to carry foreign DNA into a host cell (commonly a plasmid or virus)

- **Plasmid** — A small, circular DNA molecule found in bacteria; commonly used as a cloning vector

PCR (Polymerase Chain Reaction)

- **PCR** — A technique that amplifies (copies) a specific DNA segment millions of times in vitro
- **Denaturation (~95°C)** — Heat separates the two DNA strands
- **Annealing (~55°C)** — Short DNA primers bind to complementary sequences flanking the target region
- **Extension (~72°C)** — Taq DNA polymerase synthesizes new DNA strands from the primers
- **Taq Polymerase** — A heat-stable DNA polymerase from *Thermus aquaticus* (survives the 95°C denaturation step)
- **Primer** — A short synthetic DNA sequence that tells polymerase where to start copying

Gel Electrophoresis

- **Gel Electrophoresis** — Technique that separates DNA fragments by size using an electric field through an agarose gel
- **DNA migrates toward the positive electrode** — Because DNA is negatively charged (phosphate backbone)
- **Smaller fragments travel farther** — They move more easily through the gel pores
- **DNA Ladder** — A set of DNA fragments of known sizes used as a reference standard

Genomics & Bioinformatics

- **Genomics** — The study of entire genomes (all the DNA in an organism)
- **Human Genome Project** — Completed in 2003; mapped the ~3 billion base pairs of human DNA
- **Functional Genomics** — Determining what each gene does through gene expression patterns and knockout experiments
- **Bioinformatics** — Using computational tools to store, analyze, and interpret biological data

- **STR (Short Tandem Repeat)** — Highly variable DNA regions where a short sequence repeats; used in genetic fingerprinting

Genetic Engineering & Gene Therapy

- **Transgenic Organism (GMO)** — An organism that has received genes from another species through genetic engineering
- **Transformation** — The process of introducing foreign DNA (on a plasmid) into a bacterial cell
- **Gene Therapy** — Treating genetic disorders by introducing a functional copy of a defective gene into a patient's cells
- **Ex Vivo Gene Therapy** — Cells are removed from the patient, genetically modified in the lab, and returned
- **In Vivo Gene Therapy** — A vector delivers the therapeutic gene directly into the patient's body
- **CRISPR-Cas9** — A precise gene-editing tool that can cut DNA at a specific sequence, allowing genes to be deleted, corrected, or inserted
- **Somatic Gene Therapy** — Modifying genes in body cells (changes not passed to offspring)
- **Germline Modification** — Modifying genes in egg, sperm, or embryo cells (changes ARE inherited — raises ethical concerns)

Biotechnology Techniques: Step by Step

Creating Recombinant DNA

1. **Cut** both the donor DNA and the plasmid with the same restriction enzyme → produces matching sticky ends
2. **Mix** the fragments together → sticky ends base-pair by complementarity
3. **Seal** with DNA ligase → recombinant plasmid is complete
4. **Transform** bacteria with the recombinant plasmid → bacteria now carry the foreign gene
5. **Select** for bacteria containing the plasmid (e.g., antibiotic resistance marker)

PCR Cycle (Repeated 25-35 Times)

Step	Temperature	What Happens
Denaturation	~95°C	DNA strands separate
Annealing	~55°C	Primers bind to target sequences
Extension	~72°C	Taq polymerase builds new strands

After n cycles: **2^n copies** of the target DNA. After 30 cycles: ~1 billion copies from a single molecule.

Gel Electrophoresis

1. DNA samples loaded into wells at the **negative end** of the gel
2. Electric current applied → DNA migrates toward **positive end**
3. Smaller fragments move farther; larger fragments stay near the top
4. Bands visualized with UV light and a DNA-binding dye
5. Compare to a **DNA ladder** to estimate fragment sizes

Applications of Biotechnology

Application	Technique Used	Example
Forensic ID	STR profiling + gel electrophoresis	Crime scene DNA matching
Paternity testing	STR analysis	Child's bands must come from one parent
Medical insulin	Recombinant DNA + bacteria	Human insulin gene expressed in <i>E. coli</i>
Golden Rice	Transgenic crop engineering	Beta-carotene gene inserted into rice
Gene therapy	Viral vectors deliver functional genes	Treating cystic fibrosis, SCID
CRISPR editing	Cas9 protein + guide RNA	Correcting sickle cell mutations

Ethical Considerations

- **GMO safety** — Extensive testing required; concerns about ecological impact and consumer choice
- **Gene therapy risks** — Immune reactions to viral vectors; difficulty targeting the right cells
- **Germline editing** — Permanent, heritable changes raise concerns about unintended consequences and equity
- **Genetic privacy** — DNA databases and forensic profiling raise questions about consent and data security

Study Tips

1. **Know the tools** — Restriction enzymes cut, ligase joins, polymerase copies, electrophoresis separates
2. **PCR = molecular photocopier** — Three temperatures, three steps, exponential amplification
3. **Gel electrophoresis** — Small = fast = far from the well; always runs toward positive
4. **STRs for forensics** — Everyone has unique repeat lengths; child inherits one set from each parent
5. **Recombinant DNA process** — Cut → Mix → Ligate → Transform → Select
6. **CRISPR vs. restriction enzymes** — CRISPR edits in place with precision; restriction enzymes only cut at fixed sites
7. **Somatic vs. germline** — Somatic changes die with the patient; germline changes are inherited forever